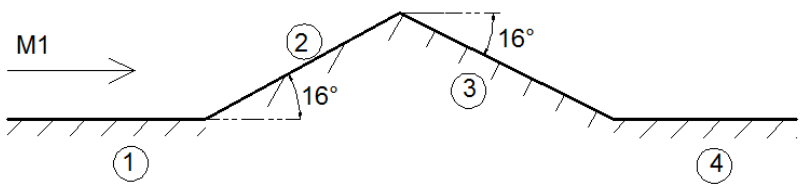
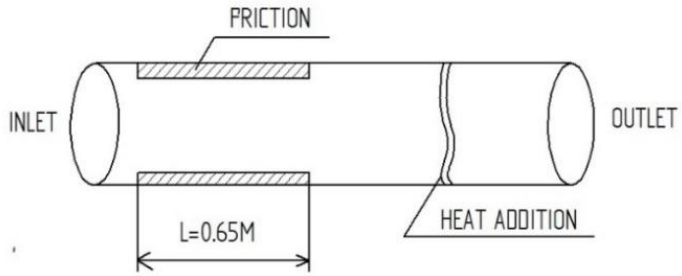
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**AERONAUTICAL**	**Course Type**	**C**
Course Name	**COMPRESSIBLE AERODYNAMICS**	**Course Code**	**AAE 3122**
Semester	**V**	**Max Marks**	**50**
Date	**23-11-2024**	**Duration**	**3 HOURS**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Series of flat plates arranged as shown in the Figure 1. Consider that a supersonic flow is passing over plate 1 with following parameters, M_1=2, P_1=1.2atm, T_1=315K, density = 1.05kg/m^3. Find the rest of flow properties a) Mach number, density , pressure, temperature on plate 3 b) Total pressure and total temperature on plate 4  Figure - 1	**5**	3	1,2,4	3	1,2,4	4
1B	How do the properties of a normal shock wave differ from those of an oblique shock wave?, and how does the each type of shock wave affect the flow properties (such as pressure, temperature and velocity)	**3**	4	1,3	4	1,3,4	3
1C	Why the linearized perturbation velocity potential equation cannot be applied to a blunt body? Also, outline the limitations of this equation.	**2**	5	1,2	5	1,2	3

2A	Consider a constant area duct with diameter of 0.32m (Figure – 2) and the inlet conditions are M_1=2, P_1=1.25atm, T_1=275K. In the duct at the first section we are having constant friction coefficient (f=0.025) and the remaining duct we adding heat q=10^5 J/Kg. If then calculate the flow properties (M, T, P, ρ and P_0) at each sections.  Figure - 2	**5**	4	1,2,4	4	1,2,3	4
2B	Differentiate how a Pitot tube is used to measure velocity in subsonic and supersonic flows, explaining each step clearly.	**3**	2	2,4	2	1,4	3
2C	Define a supercritical airfoil and explain how it differs from conventional airfoils.	**2**	4	3,4	4	3,4	2
3A	Consider a horizontal supersonic flow at Mach 2.58 with a static pressure & temperature of 1.2atm & 290K respectively. This flow passes over a compression corner with deflection angle of 15 deg. Once the shock is generated it's interacting with the solid wall just above the compression corner. If then, calculate the reflected angle made by the reflected shock with respect to the solid wall and Mach number, pressure & temperature behind the reflected shock.	**5**	3	2,4	3	1,3	3
3B	Consider the purely subsonic flow in a convergent divergent duct. The inlet, throat and exit are 1m^2, 0.7m^2, and 0.85m^2, respectively. If the inlet Mach number and pressure are 0.3 and 80000 N/m^2 respectively. Calculate a) Mach number and Pressure at the throat b) Mach number and Pressure at the exit	**3**	5	2,4	5	1,3	3
3C	Define compressibility correction and illustrate it with an example.	**2**	5	4	5	3	2
4A	Consider an aircraft with planform area 28m^2 in steady level flight at Mach 2.56 at altitude of 11km. The weight of the aircraft is 12556kgf. Assume that all the lift of the airplane comes from the lift on the wings also assuming the lift coefficient of the wing is the same as the lift coefficient for the airfoil section. The chord length of the airfoil is 2.8m which almost equal to the mean chord length of the wing. If then, calculate the followings ($\rho_\infty = 0.3648\frac{kg}{m^3}, T_\infty = 216.78K$) a) Calculate the angle of attack with the help of linearized equations b) If it is fully laminar flow calculate the coefficient of friction and wave drag c) At what angle of attack C_f is equal to C_D?	**5**	5	2	5	2	3

4B	Describe the followings: a) Area rule b) Why we use conical nose section in supersonic aircrafts? c) Why in modern aircrafts use thin airfoils?	**3**	2	1,4	2	1,4	2
4C	Describe how to determine the wave angle through the shock polar and illustrate with a diagram.	**2**	3	1,2	3	1,4	3
5A	Consider a rocket engine burning hydrogen and oxygen in the combustion chamber and here the temperature and pressure are 3600K and 28atm respectively. The molecular weight of the chemically reacting gas in the combustion chamber 18 and $\gamma = 1.22$. The pressure at the exit of the convergent – divergent rocket nozzle is 1.31×10^2 atm. The area of the throat is 0.42 m^2. Assuming a calorically perfect gas. Calculate : a) The exit Mach number b) Exit velocity c) The mass flow through the nozzle d) Area at the exit	**5**	4	1,4	4	1,4	3
5B	In a supersonic wind tunnel how the diffuser efficiency is related to diffuser throat area (explain it through graphical representation)? Also write down how to eliminate the initial shock wave generated inside the wind tunnel.	**3**	3	1,3	3	1,3	3
5C	A blunt nosed missile is flying at Mach 1.6 at standard sea level. Calculate the temperature and pressure at the nose of the missile.	**2**	4	2	4	2	4

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	
2	***3+3***
3	***2+5+5+3***
4	***2+2+3+5+5***
5	***2+2+3+5***

BL	Total Marks
2	**2+2+3**
3	**2+2+3+3+5+3+5+5+3**
4	**2+5+5**
5	